Geometry & Measurement

Section 1: Angles & Triangles

1 In a triangle, the angles are in the ratio $2:3:5$. What is the measure of the largest angle?

a. 90°

b. 100°

c. 80°

d. 75°

2 In a right triangle, one leg is $12\ \text{cm}$ and the hypotenuse is $13\ \text{cm}$. What is the area of the triangle?

a. $30\ \text{cm}^2$

b. $60\ \text{cm}^2$

c. $39\ \text{cm}^2$

d. $25\ \text{cm}^2$

3 Two parallel lines are intersected by a transversal. If an interior angle measures $(3x - 15)^\circ$ and its consecutive interior angle measures $(2x + 20)^\circ$, what is x?

a. 35

b. 30

c. 40

d. 25

Section 2: Circles, Circumference & Area

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4 A circle has a circumference of 16π cm. What is its area?

a. 64π cm^2

b. 32π cm^2

c. 16π cm^2

d. 128π cm^2

5 A square of side length 10 cm has a circle inscribed within it. What is the area of the shaded region between the square and the circle?

a. $100 - 25\pi$

b. $100 - 10\pi$

c. $50 - 25\pi$

d. $100 - 50\pi$

6 If the radius of a circle is doubled, by what factor does its area increase?

a. 4

b. 2

c. 8

d. 16

Section 3: Quadrilaterals & Polygons

7 A rectangle has a perimeter of 40 cm and its length is 3 times its width. What is the area of the rectangle?

a. 75 cm^2

b. 80 cm^2

c. 100 cm^2

d. 60 cm^2

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8 What is the sum of the interior angles of a regular hexagon?

a. 720°

b. 540°

c. 360°

d. 900°

9 A rhombus has diagonals of lengths $12\ \text{cm}$ and $16\ \text{cm}$. What is the length of one side of the rhombus?

a. $10\ \text{cm}$

b. $8\ \text{cm}$

c. $14\ \text{cm}$

d. $20\ \text{cm}$

Section 4: Solid Geometry & Volume

10 A rectangular tank has dimensions $4\ \text{m} \times 5\ \text{m} \times 6\ \text{m}$. If it is filled with water at a rate of $10\ \text{m}^3/\text{min}$, how many minutes does it take to fill completely?

a. 12 minutes

b. 10 minutes

c. 15 minutes

d. 20 minutes

11 A cylinder has base radius $3\ \text{cm}$ and height $10\ \text{cm}$. What is its volume in cubic centimeters?

a. 90π

b. 60π

c. 30π

d. 180π

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12 If the volume of a cube is $64\ cm^3$, what is its total surface area?

a. $96\ cm^2$

b. $64\ cm^2$

c. $128\ cm^2$

d. $48\ cm^2$

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